CBAM Corridor Cost Model

Status Report

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Date: 27 July 2026

Purpose: Briefing for supervisor meeting. Summarises what has been built, what is assumed rather than confirmed, what remains outstanding, and recommended next steps.

1. What the model does

The model prices the carbon compliance cost of importing hydrogen and ammonia along two maritime corridors under two different regulatory regimes: Halifax to Hamburg, which falls under EU CBAM (Carbon Border Adjustment Mechanism, live since 1 January 2026), EU ETS Maritime, and FuelEU Maritime; and Ningbo to Felixstowe, which falls under UK CBAM (not starting until 1 January 2027) and UK ETS Maritime. The two regimes running on different timelines is the central finding the dissertation is built around, not an inconvenience to model past.

It works in two layers. The maritime layer, built from Gayu's shipping data, prices carbon cost per voyage. The CBAM layer, built from Riya's emissions data, prices the border tax per tonne of product. A cargo capacity figure from Gayu converts between the two, so the model can report total carbon compliance cost per tonne of hydrogen or ammonia delivered, in the corridor's own currency.

The codebase is roughly 3,200 lines of Python, organised into a regulatory constants module, four cost calculation modules (CBAM, maritime ETS, FuelEU, and the layer that joins them), a data loading and validation layer, a scenario runner, and a sensitivity analysis module. It is tested by 81 automated tests and runs end to end from a single notebook, run_model.ipynb.

2. Errors caught in the original build specification

Three errors in the project's original technical build specification were found and corrected before they reached results.

The FuelEU Maritime penalty formula was missing a divisor. The specification calculated the penalty without dividing by the ship's actual fuel intensity, which Annex IV Part B of Regulation (EU) 2023/1805 requires. This would have overstated the penalty by roughly ninety times. Verified directly against the regulation, and independently confirmed by Gayu's separate implementation, which used the correct formula without prompting.

The origin carbon price adjustment had a units error. The specification subtracted a price per tonne of CO_2 directly from a total cost figure, which could produce a negative CBAM liability from a positive one under realistic inputs. Corrected so the adjustment scales with the same emissions volume and CBAM factor as the liability it offsets, consistent with Article 9 of Regulation (EU) 2023/956.

The FuelEU target schedule stopped at 2029, while the scenario matrix runs through 2030, where Article 4(2) of the regulation tightens the reduction requirement from 2% to 6%. The full step schedule through 2050 has been added.

Gayu's maritime notebooks also corrected the build specification on several points, most significantly the Halifax to Hamburg distance, which the specification listed as approximately 6,300 nautical miles against the correct SeaRoute derived figure of 2,962 nautical miles. This had been overstating that corridor's voyage emissions by more than a factor of two.

3. What was assumed, and what has since been resolved

This is the section that matters most. Every input the model depends on is listed below with its current status.

Input	Status	Detail
Maritime costs (distance, fuel, vessel specs)	**Resolved**	From Gayu's three notebooks. All 31 published figures reproduce exactly in the model's test suite.
Cargo tonnage per voyage	**Resolved**	From Gayu's cargo capacity notebook: 84,000 m3 vessel (Seo et al., 2024), 98% IMO filling limit, giving 56,142 t ammonia or 5,828 t hydrogen. See caveat below.
Hydrogen embedded emissions	**Resolved**	From Riya, delivered 26 July 2026 and revised 4 August 2026, both corridors, four pathways each, all cited to named sources. The 4 August revision moved Canada blue from 4.89 to 2.02. All three China hydrogen pathways were then consolidated onto a single study (S0360319925010602), agreed with Riya, replacing figures that had been drawn from four different papers: grey 29.02 to 20.09 and blue 7.91 to 6.28.
Ammonia embedded emissions	**Delivered**	Delivered 29 July 2026 and revised 4 August 2026. Both corridors, with CBAM regulatory defaults (1.98 Canada, 4.36 China). The 4 August revision corrected China coal gasification from 4.60 to 6.15: the old value was the same paper's natural-gas row, mislabelled.
UK ETS maritime price	**Resolved**	GBP 49.41/tCO2e, the UK ETS Authority's official determination for the 2026 scheme year, published 28 November 2025. Calculated from twelve months of UKA December futures settlement prices, so market derived rather than an administrative figure.
EU ETS price	**Resolved**	Two anchor years used: 2026 from ESMA's near term market range (via Gayu), 2030 from a multi institution consensus forecast. The model interpolates between them by year.
Canada origin carbon price	**Resolved, with a caveat**	CAD 110/tCO2e, the federal Output Based Pricing System rate for 2026 (Nova Scotia, where EverWind is based, follows the federal rate directly). Converted to EUR 68.63 at the ECB reference rate for 23 July 2026. Caveat: this system only charges on emissions above a facility specific benchmark, so the true figure paid could be lower. Treat as an upper bound, not a confirmed effective price.
China origin carbon price	**Resolved**	Set to EUR 0. This is a finding, not a gap: hydrogen production is not currently within the scope of China's national Emissions Trading Scheme, which as of 2026 covers only power generation, steel, cement, and aluminium. Chemicals and petrochemicals, which would include hydrogen, are described in official and industry sources as planned for a future expansion phase.
UK CBAM phase in mechanism	**Resolved**	Traced through primary UK legislation: the draft CBAM (Calculation of CBAM Rate and Determination of Carbon Price Relief) Regulations 2026 and the Greenhouse Gas Emissions Trading Scheme (Amendment) Order 2026. The UK CBAM rate is not a flat percentage of embedded emissions like the EU's; it is UK ETS price times one minus a baseline free allocation percentage times an Article 16(14) factor that runs 0.975 in 2027 down to 0.775 by 2030. The baseline (Finance Act 2026 s.149(4)) blends 2019 EU ETS data with 2022 and 2023 UK ETS data for the UK's one in scope hydrogen installation, Teesside Hydrogen Plant. All three years are now sourced, giving a baseline of 86.49% and an implied rate rising from 15.7% of the UK ETS price in 2027 to 33.0% by 2030.

Input	Status	Detail
GBP to EUR exchange rate	**Resolved**	ECB reference rate for 23 July 2026, 1 GBP = 1.17209 EUR, the same reference date used for the Canadian dollar and US dollar conversions. Headline tables still report each corridor in its own currency, as Gayu's notebooks do; the rate is applied only where a single-currency comparison is explicitly labelled as such.
Production cost	**Resolved**	From Riya, 4 August 2026, covering every modelled pathway on both corridors. Converted from USD at the 23 July 2026 ECB rate. Her two sheets disagree on China green hydrogen (4.63 vs 5.72-6.62 USD/kg); the lower figure is used and the choice needs stating in the methodology.
Conversion and shipping cost	**Not sourced, no owner**	See Section 4. Both are invariant to production pathway, so they cancel out of within-corridor pathway comparisons and do not block the marginal abatement cost results.

4. What is still missing

Production, conversion, and freight cost per tonne of product. This is the most significant open gap. The model currently produces total carbon compliance cost, meaning CBAM plus maritime ETS plus FuelEU. It does not yet produce a total delivered cost, because three of the six cost components that would make up a delivered cost figure have never been assigned to a member of the group. Production cost in particular is understood from the literature to be the largest single component of delivered cost, larger than any of the regulatory charges the model already covers. Until this is resolved, the dissertation's aim should be described as pricing regulatory compliance cost rather than total delivered cost, unless the wording of the aims section already reflects this distinction.

Ammonia embedded emissions, pending Riya.

Two provisional emissions figures from Riya, Canada green and blue hydrogen, pending her confirmation.

China blue hydrogen production method. Riya's data gives an emissions figure for this pathway but does not state the production method. It has been labelled generically pending confirmation, since China's blue hydrogen route is more likely coal based with carbon capture than the gas based route used for Canada.

5. Quality control performed

Three layers of checking are built into the model, rather than trust being placed in a single run.

Eighty one automated tests check the regulatory logic against hand calculated expected values, including specific tests for the two facts previously confirmed wrong earlier in this project: the CBAM factor being confused with the free allocation share, and UK ETS maritime scope being assumed equivalent to the EU's when it is not.

A reproduction test suite checks the model against all 31 figures published in Gayu's maritime notebooks. All 31 currently reproduce exactly.

An external cross check was attempted against Ramsook, Boodlal and Maharaj (2025), a published study of Trinidad and Tobago's ammonia exports under EU CBAM. This check does not currently reconcile: the model produces a substantially higher CBAM burden as a share of export value than the 22% reported in that paper. The most likely explanation, not yet confirmed, is that the published figure is measured against total global export revenue while this model's CBAM cost applies only to the EU bound share. This is reported honestly in the model's output rather than adjusted until it agrees, and is flagged as an open item requiring the source paper to be read in full.

6. Headline results so far

Using currently available data, 2026 shows a pronounced asymmetry: Chinese hydrogen, despite having roughly twice the embedded emissions of Canadian hydrogen, faces a carbon compliance cost roughly 37 times lower, because UK CBAM has not yet started and UK ETS does not price the ocean voyage. By 2030, this narrows substantially and can invert, as the EU CBAM factor rises sharply and UK CBAM (modelled here as an upper bound scenario, given the unresolved phase in question) begins to apply. The asymmetry therefore appears to be a temporary feature of the current regulatory timeline rather than a structural one, which is a stronger and more defensible framing for the discussion chapter than a simple statement of which corridor is cheaper.

7. Recommended next steps

1. Decide ownership of production, conversion, and shipping cost this week. Either assign it or formally narrow the dissertation's stated aim to compliance cost rather than delivered cost.
2. Follow up with Riya on the ammonia table and the two provisional Canadian hydrogen figures.
3. Continue monitoring UK CBAM policy publications for the free allowance adjustment mechanism referenced in paragraph 3.76 of the government's consultation response, and update the model once it is published.
4. Resolve the Ramsook et al. reference check by reading the source paper in full to confirm or rule out the denominator hypothesis.
5. Await MCG's HyPACT validation data, requested from Clinton and Jiaqi, subject to ethics approval sign off.